

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Data Warehousing and Data Mining

COURSE CODE: CSA1603

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 20%

QUESTIONS: 10

Total Marks:20

Annexure A

CLASS TEST

<i>Criteria</i>	Understanding of Task (2)	Procedure & Execution (2.5)	Observation & Result (2)	Viva Voce / Conceptual Response (2)	Time Management & Presentation (1.5)	<i>Total(10)</i>
<i>Weightage</i>	20%	25%	20%	20%	15%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						
<i>Q6</i>						
<i>Q7</i>						
<i>Q8</i>						
<i>Q9</i>						
<i>Q10</i>						

Code of Conduct:

I B.AVINASH(192425373)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/ experiment without assistance	Understands task with minor clarification	Partial understanding ; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
V i v a Vo c e / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

CLASS TEST

QUESTIONS:

Total Marks: 10x2 =20

Q1. Define Data Mining and list any two basic data mining tasks. Explain the steps involved in KDD process and justify the role of data preprocessing in it. [L2, L4]

Q2. What is data cleaning? Mention two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples. [L1, L2, L3]

Q3. Define data integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration. [L2, L4]

Q4. What is covariance analysis? List the two covariance formulas [L2, L3]

Q5. Suppose two stocks A and B have the following values in one week: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). Question: If the stocks are affected by the same industry trends, will their prices rise or fall together and independent ?

Q6. Explain the architecture of a typical data mining system with a neat diagram. Discuss the role of each component and analyze how they interact to support knowledge discovery. [L2, L4]

Q7. Find covariance for following data set $x = \{4,5,6,7,9\}$, $y = \{8,3,7,5,6\}$

Q8. Describe the major steps in data pre-processing. Given a dataset with missing values, outliers, and inconsistent formats, propose a systematic pre-processing pipeline and justify the sequence of steps. [L3, L4, L5]

Q9. A hospital is analyzing patient **age** data to identify broad health trends. Because the raw data contains minor variations and noise, the analyst needs to perform local smoothing using **equal-frequency** (equi-depth) binning. **Sample Data Set (Ages):** 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 [L2, L4]

Q10. Explain data discretization and its importance in data mining. Compare equal-width and equal-frequency binning with suitable examples, and evaluate their impact on data analysis. [L3, L4, L5]

1. Data Mining is the process of extracting useful patterns, trends and knowledge from large volumes of data using statistical and machine learning techniques.

Two basic data mining tasks:

Classification → Assigns data into predefined categories

Clustering → Groups similar data objects without predefined labels.

2. KDD (Knowledge Discovery in Databases) is the process of discovering useful knowledge from data.

Steps of KDD:

- Data cleaning
- Data integration
- Data selection
- Data Mining
- Pattern Evaluation

Importance of Data preprocessing

- Removes errors and missing values.
- Improves the accuracy and efficiency of data mining.

3. Data Cleaning is the process of detecting and correcting inaccurate, incomplete or inconsistent data.

Two data quality issues:

- Missing values
- Noisy (incorrect) data

4. Missing values:

- Replace with mean, median or mode.
- Predict values using machine learning methods.

Noisy data:

- Binning
- Regression
- Clustering

These methods improve data quality before mining.

5. Data Integration is the process of combining data from multiple sources into a single, consistent database.

Challenges:

- Data redundancy (duplicate data)
- Data inconsistency (different format or values)

6. Covariance measures the relationship b/w variables and indicates whether they move together.

population Covariance:

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{N}$$

Sample Covariance:

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

7. A typical Data mining system consists of:

- Data base / Data warehouse
- Data cleaning and integration
- Data mining engine
- pattern Evaluation module.

It converts raw data into useful knowledge.

8. The major steps are:

- Data cleaning
- Data integration
- Data Transformation
- Data Reduction

These steps improve data quality and reduce processing time

9. Equal-Frequency Binning is a discretization method in which each bin contains an equal number of data values.

Advantages:

- Reduces noise
- Handles skewed data effectively
- Improves data analysis.

10. Data Discretization is the process of converting continuous data into discrete intervals (bins)

Equal-Width Binning

equal interval size

Simple to implement

Sensitive to outliers

Equal-Frequency Binning

Equal number of values

Better for skewed data

Less affected by outliers